Constraining Dust Attenuation in Star Forming Galaxies Using JWST

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INTRODUCTION

This research establishes a framework for robustly measuring dust attenuation during the epoch of peak cosmic star formation ($1.5 \leq z \leq 3.5$).

- **Why Dust Matters:** Interstellar dust absorbs and scatters rest-frame optical/UV tracers. Correcting for this obscuration is one of the largest systematic uncertainties in measuring intrinsic Star Formation Rates (SFRs).
- **The Problem:** Traditional Balmer decrement ($H\alpha/H\beta$) methods are fundamentally degenerate; they can constrain total attenuation but force an assumption regarding the shape of the dust curve.
- **The Solution (Paschen Lines):** Deep JWST NIRSpec spectroscopy allows simultaneous access to rest-frame optical (Balmer) and near-infrared (Paschen) emission lines. Because Paschen lines are at longer wavelengths and less obscured by dust, this wide baseline breaks the degeneracy, allowing us to simultaneously constrain both the dust curve slope and normalization.

GOVERNING EQUATIONS

We characterize the wavelength dependence of the effective optical depth using a power-law form, where n is the slope and τ_V is the normalization:

$$\tau(\lambda) = \tau_V \left(\frac{\lambda}{5500\text{Å}}\right)^{-n}$$

Applying these dust constraints allows us to recover intrinsic Star Formation Rates. Our uncorrected baselines utilize the standard calibrations from Kennicutt & Evans (2012) and Cleri et al. (2022):

$$\text{SFR}_{H\alpha}[M_\odot \text{yr}^{-1}] = 5.37 \times 10^{-42} L_{H\alpha}[\text{erg s}^{-1}]$$
$$\text{SFR}_{Pa\beta}[M_\odot \text{yr}^{-1}] = 10^{-40.02} L_{Pa\beta}[\text{erg s}^{-1}]$$

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HIERARCHICAL AGN REMOVAL PIPELINE

Accurate dust attenuation modeling requires a purely star-forming sample. Because our method assumes intrinsic line ratios are governed strictly by Case B recombination in H II regions, any contamination from Broad-Line Regions (BLR) or Narrow-Line Regions (NLR) in Active Galactic Nuclei (AGN) will artificially skew observed line ratios and invalidate the dust models.

- **Hard Indicators (AGN):** Excluded if mid-infrared fraction $f_{AGN} > 0.30$ (via AGNBoost) or X-ray luminosity $L_X > 10^{42}$ erg s^{-1}.
- **Soft Indicators (AGN Candidates):** Excluded if exhibiting *both* broad-line kinematics (FWHM > 1000 km s^{-1}) and AGN-like optical line ratios on the OHNO diagram.
- **Final Counts:**
  - **SFG:** 4061 Galaxies
  - **AGN:** 202 Galaxies
  - **AGN Candidate:** 587 Galaxies

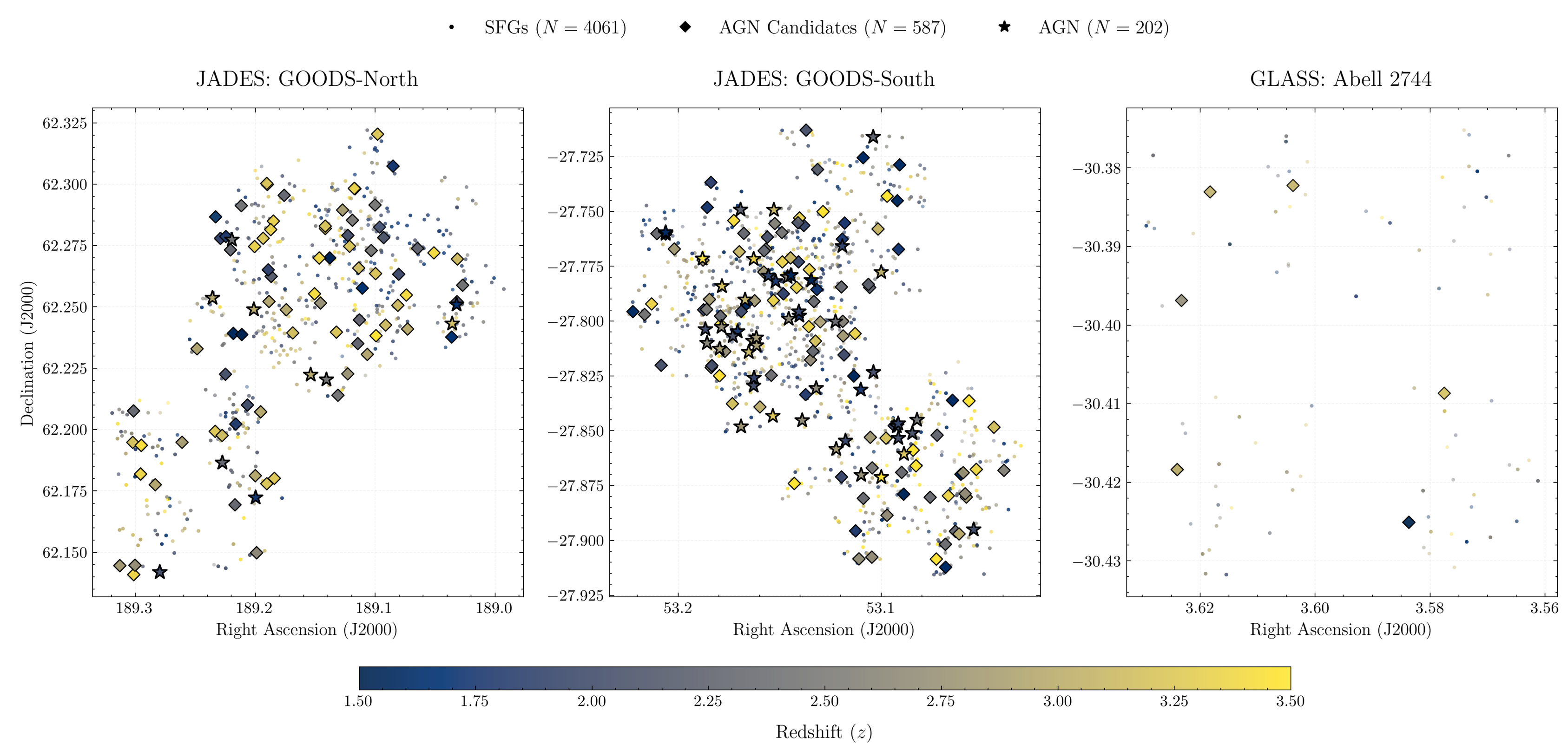

Figure 1: Spatial distribution and final boolean classification of the sample across GOODS-North, GOODS-South, and Abell 2744.

RESULTS & CONCLUSIONS

The Multi-Line Method

For each pristine galaxy, we utilize measurements of $H\alpha$, $H\beta$, and a Paschen transition ($Pa\beta$ or $Pa\gamma$). A Newton-Raphson numerical solver iteratively determines the effective attenuation parameters (τ_V, n) that align predicted line ratios with observed values.

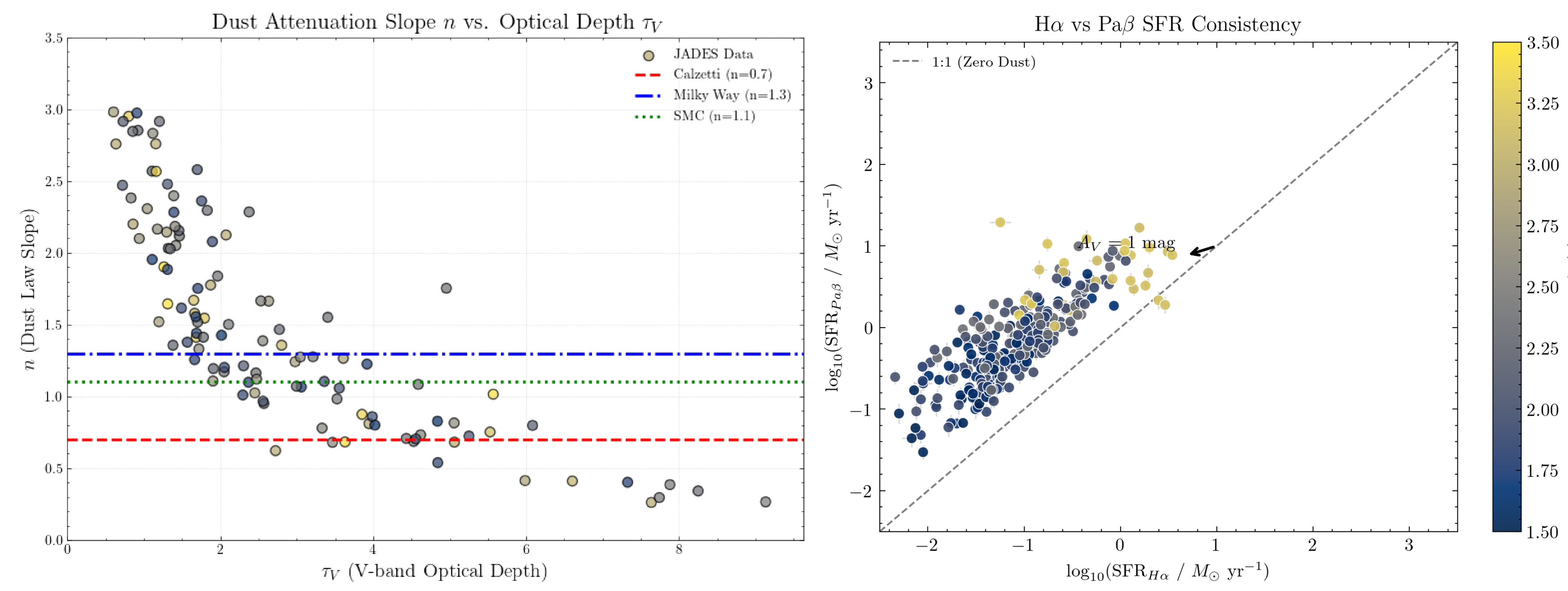

Figure 2: (Left) Comparison of inferred dust attenuation slope (n) vs. optical depth (τ_V) with overlaid constant Balmer decrement lines. (Right) Before vs. After SFR comparison demonstrating the recovery of intrinsic SFRs.

Conclusions

- **Degeneracy Broken:** Combining Balmer and Paschen lines across a wide baseline successfully breaks the degeneracy between dust normalization and slope.
- **SFR Recovery:** Multi-line corrections resolve discrepancies between optical and near-IR tracers, demonstrating the severe impact of dust at Cosmic Noon.

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